Sri Lanka Institute of Information Technology

IT3010 – Network Design and Management

Lab Sheet 5 – Lab Report

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1. Introduction

Simple Network Management Protocol (SNMP) is widely used to monitor and manage network devices like routers, switches, and servers. It allows network administrators to gather and organize data from devices on an IP network, and also enables remote configuration changes when needed.

In this hands-on session, SNMP was installed and set up on a Linux server. Various command-line tools—such as snmptranslate, snmpget, and snmpgetnext—were then used to access and retrieve information stored in the Management Information Base (MIB).

2. Procedure

STEP 01 – Stop Running Services

service dhcpd stop

service named stop

Figure 1 stop running service

service dhcpd status

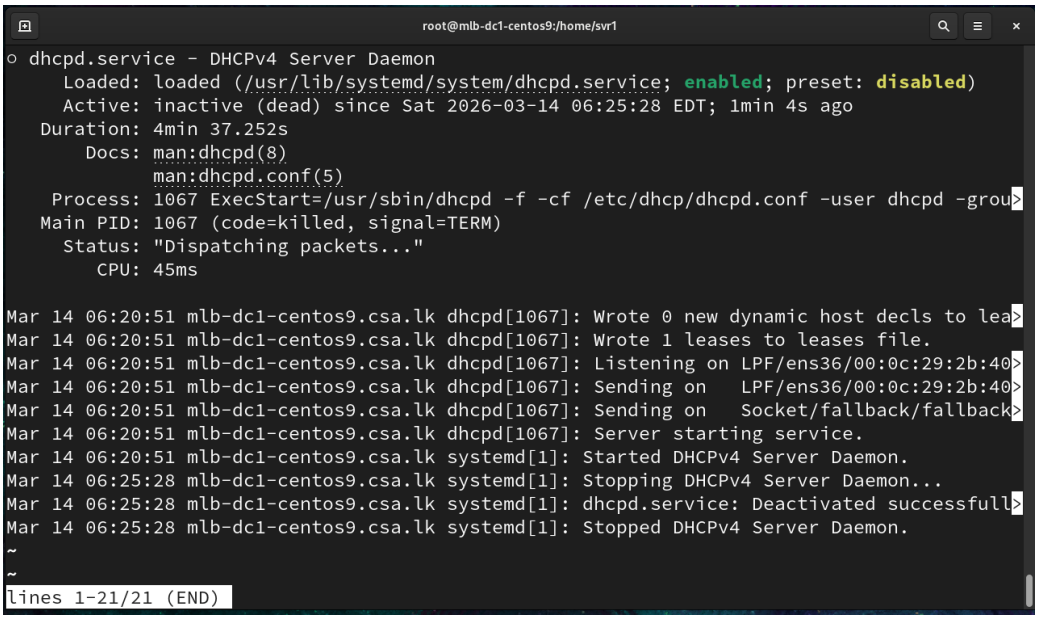

Figure 2 Check the DHCPD Status

service named status

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# service named status
Redirecting to /bin/systemctl status named.service
○ named.service - Berkeley Internet Name Domain (DNS)
     Loaded: loaded (/usr/lib/systemd/system/named.service; disabled; preset: disabled)
     Active: inactive (dead)
[root@mlb-dc1-centos9 svr1]#
```

Figure 3 Check the Status

ifconfig

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.234.128  netmask 255.255.255.0  broadcast 192.168.234.255
        inet6 fe80::20c:29ff:fe2b:4013  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:2b:40:13  txqueuelen 1000  (Ethernet)
        RX packets 883325  bytes 1283865808 (1.1 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 114905  bytes 6938194 (6.6 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ens36: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.1.2  netmask 255.255.255.0  broadcast 10.0.1.255
        inet6 fe80::1d81:51ca:cc03:340c  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:2b:40:1d  txqueuelen 1000  (Ethernet)
        RX packets 49  bytes 4547 (4.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 43  bytes 4733 (4.6 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 17  bytes 2099 (2.0 KiB)
```

Figure 4 Check IP addresses

STEP 02 – Install SNMP Packages

yum install –y net-snmp net-snmp-utils

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# yum install -y net-snmp net-snmp-utils
CentOS Stream 9 - BaseOS                                 3.7 kB/s | 5.0 kB     00:01
CentOS Stream 9 - BaseOS                                 1.2 MB/s | 8.9 MB     00:07
CentOS Stream 9 - AppStream                              7.4 kB/s | 5.0 kB     00:00
CentOS Stream 9 - AppStream                              2.4 MB/s |  27 MB     00:11
CentOS Stream 9 - Extras packages                         13 kB/s | 5.0 kB     00:00
CentOS Stream 9 - Extras packages                        1.2 kB/s |  20 kB     00:17
Last metadata expiration check: 0:00:02 ago on Sat 14 Mar 2026 06:40:19 AM EDT.
No match for argument: -y
Package net-snmp-1:5.9.1-21.el9.x86_64 is already installed.
Package net-snmp-utils-1:5.9.1-21.el9.x86_64 is already installed.
Error: Unable to find a match: -y
[root@mlb-dc1-centos9 svr1]#
```

Figure 5 Install SNMP Packages

STEP 03 – Configure Static IP with Host-Only Network

Manually configure the server network settings as follows:

IP Address: 10.0.1.2
Netmask: 255.255.255.0
Gateway: 10.0.1.1

vi /etc/sysconfig/network-scripts/ifcfg-ens36

```
root@mlb-dc1-centos9:/home/svr1
IPADDRESS = 10.0.1.2
NETMASK=255.255.255.0
GATEWAY=10.0.1.1
~
~
~
```

Restart Network

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# systemctl restart NetworkManager
[root@mlb-dc1-centos9 svr1]#
```

Check Status

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# systemctl restart NetworkManager
[root@mlb-dc1-centos9 svr1]# systemctl status NetworkManager
● NetworkManager.service - Network Manager
     Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled; preset: ena>
     Active: active (running) since Sat 2026-03-14 06:57:27 EDT; 54s ago
       Docs: man:NetworkManager(8)
   Main PID: 2750 (NetworkManager)
      Tasks: 3 (limit: 10717)
     Memory: 6.9M
        CPU: 408ms
     CGroup: /system.slice/NetworkManager.service
             └─2750 /usr/sbin/NetworkManager --no-daemon

Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9004] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9106] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9147] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9231] ma>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9348] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9391] ma>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9521] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9612] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9700] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9741] ma>
lines 1-21/21 (END)
```

STEP 04 – Configure SNMP

vi /etc/snmp/snmpd.conf

```
root@mlb-dc1-centos9:/home/svr1
# enterprises.ucdavis.255.5 = 42
# enterprises.ucdavis.255.6 = Gauge: 42
#
# % snmpget -v 1 localhost public .1.3.6.1.4.1.2021.255.5
# enterprises.ucdavis.255.5 = 42
#
# % snmpset -v 1 localhost public .1.3.6.1.4.1.2021.255.1 s "New string"
# enterprises.ucdavis.255.1 = "New string"
#

# For specific usage information, see the man/snmpd.conf.5 manual page
# as well as the local/passtest script used in the above example.

###############################################################################
# Further Information
#
#  See the snmpd.conf manual page, and the output of "snmpd -H".

rocommunity  public  10.0.1.2
rocommunity  public   127.0.0.1

syslocation  "HYD, UM DataCenter"
syscontact  admin@ndm.lk
"/etc/snmp/snmpd.conf" 468L, 18988B                                  468,25        Bot
```

service snmpd start

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# service snmpd start
Redirecting to /bin/systemctl start snmpd.service
[root@mlb-dc1-centos9 svr1]# service snmpd status
Redirecting to /bin/systemctl status snmpd.service
● snmpd.service - Simple Network Management Protocol (SNMP) Daemon.
     Loaded: loaded (/usr/lib/systemd/system/snmpd.service; disabled; preset: disabled)
     Active: active (running) since Sat 2026-03-14 07:32:15 EDT; 5s ago
   Main PID: 3183 (snmpd)
      Tasks: 1 (limit: 10717)
     Memory: 9.9M
        CPU: 201ms
     CGroup: /system.slice/snmpd.service
             └─3183 /usr/sbin/snmpd -LS0-6d -f

Mar 14 07:32:14 mlb-dc1-centos9.csa.lk systemd[1]: Starting Simple Network Management Prot>
Mar 14 07:32:15 mlb-dc1-centos9.csa.lk snmpd[3183]: NET-SNMP version 5.9.1
Mar 14 07:32:15 mlb-dc1-centos9.csa.lk systemd[1]: Started Simple Network Management Prot>
lines 1-13/13 (END)
```

STEP 05 – Explore the MIB Tree

I. snmptranslate

1. % snmptranslate .1.3.6.1.2.1.1.3.0

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmptranslate .1.3.6.1.2.1.1.3.0
DISMAN-EVENT-MIB::sysUpTimeInstance
[root@mlb-dc1-centos9 svr1]#
```

2. % snmptranslate -On SNMPv2-MIB::system.sysUpTime.0

```
[root@mlb-dc1-centos9 svr1]# snmptranslate -On SNMPv2-MIB::system.sysUpTime.0
.1.3.6.1.2.1.1.3.0
[root@mlb-dc1-centos9 svr1]#
```

3. % snmptranslate .iso.3.6.1.private.enterprises.2021.2.1.prNames.0
   % snmptranslate -On .iso.3.6.1.private.enterprises.2021.2.1.prNames.0

```
[root@mlb-dc1-centos9 svr1]# snmptranslate .iso.3.6.1.private.enterprises.2021.2.1.prNames.0
UCD-SNMP-MIB::prNames.0
[root@mlb-dc1-centos9 svr1]# snmptranslate -On .iso.3.6.1.private.enterprises.2021.2.1.prNames.0
.1.3.6.1.4.1.2021.2.1.2.0
[root@mlb-dc1-centos9 svr1]#
```

4. % snmptranslate -Of .iso.3.6.1.private.enterprises.2021.2.1.prNames.0

```
[root@mlb-dc1-centos9 svr1]# snmptranslate -Of .iso.3.6.1.private.enterprises.2021.2.1.prNames.0
.iso.org.dod.internet.private.enterprises.ucdavis.prTable.prEntry.prNames.0
[root@mlb-dc1-centos9 svr1]#
```

5. % snmptranslate -TB 'vacm.*table'

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmptranslate -TB 'vacm.*table'
SNMP-VIEW-BASED-ACM-MIB::vacmViewTreeFamilyTable
SNMP-VIEW-BASED-ACM-MIB::vacmAccessTable
SNMP-VIEW-BASED-ACM-MIB::vacmSecurityToGroupTable
SNMP-VIEW-BASED-ACM-MIB::vacmContextTable
NET-SNMP-VACM-MIB::nsVacmAccessTable
[root@mlb-dc1-centos9 svr1]#
```

6. % snmptranslate -On -Td -Ib 'sys.*ime'

```
[root@mlb-dc1-centos9 svr1]# snmptranslate -On -Td -Ib 'sys.*ime'
.1.3.6.1.2.1.1.9.1.4
sysORUpTime OBJECT-TYPE
  -- FROM       SNMPv2-MIB
  -- TEXTUAL CONVENTION TimeStamp
  SYNTAX        TimeTicks
  MAX-ACCESS    read-only
  STATUS        current
  DESCRIPTION   "The value of sysUpTime at the time this conceptual
                 row was last instantiated."
::= { iso(1) org(3) dod(6) internet(1) mgmt(2) mib-2(1) system(1) sysORTable(9) sysOREntry(1) 4
}
[root@mlb-dc1-centos9 svr1]#
```

7. % snmptranslate -Tp -IR system

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmptranslate -Tp -IR system
+--system(1)
   |
   +-- -R-- String    sysDescr(1)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -R-- ObjID     sysObjectID(2)
   +-- -R-- TimeTicks sysUpTime(3)
   |  |
   |  +--sysUpTimeInstance(0)
   |
   +-- -RW- String    sysContact(4)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -RW- String    sysName(5)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -RW- String    sysLocation(6)
```

II. **snmpget :** retrieving data from a host.

1. % snmpget -v2c -c public 10.0.1.2 system.sysUpTime.0

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmpget -v2c -c public 10.0.1.2 system.sysUpTime.0
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (150306) 0:25:03.06
[root@mlb-dc1-centos9 svr1]#
```

2. % snmpget -v 2c -c public 10.0.1.2 system.sysUpTime.0

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmpget -v 2c -c public 10.0.1.2 system.sysUpTime.0
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (240004) 0:40:00.04
```

III. **snmpgetnext :** retrieving unknown indexed data.

1. % snmpgetnext -v 2c -c demopublic test.net-snmp.org system.sysUpTime.0
   % snmpgetnext -v 2c -c public 10.0.1.2 system.sysUpTime.0

```
[root@mlb-dc1-centos9 svr1]# snmpgetnext -v 2c -c public 10.0.1.2 system.sysUpTime.0
SNMPv2-MIB::sysContact.0 = STRING: admin@ndm.lk
[root@mlb-dc1-centos9 svr1]#
```

2. % snmpgetnext -v 2c -c demopublic test.net-snmp.org system.sysContact.0
   % snmpgetnext -v 2c -c public 10.0.1.2 system.sysContact.0

```
[root@mlb-dc1-centos9 svr1]# snmpgetnext -v 2c -c public 10.0.1.2 system.sysContact.0
SNMPv2-MIB::sysName.0 = STRING: mlb-dc1-centos9.csa.lk
[root@mlb-dc1-centos9 svr1]#
```

3. % snmpgetnext -v 2c -c demopublic test.net-snmp.org system.sysUpTime
   % snmpgetnext -v 2c -c public 10.0.1.2 system.sysUpTime

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmpgetnext -v 2c -c public 10.0.1.2 system.sysUpTime
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (278294) 0:46:22.94
[root@mlb-dc1-centos9 svr1]#
```

4. % snmpgetnext -v 2c -c demopublic test.net-snmp.org .1.3.6
   % snmpgetnext -v 2c -c public 10.0.1.2 .1.3.6

```
[root@mlb-dc1-centos9 svr1]# snmpgetnext -v 2c -c public 10.0.1.2 .1.3.6
SNMPv2-MIB::sysDescr.0 = STRING: Linux mlb-dc1-centos9.csa.lk 5.14.0-565.el9.x86_64 #1 SMP PREEM
PT_DYNAMIC Fri Feb 7 18:47:11 UTC 2025 x86_64
[root@mlb-dc1-centos9 svr1]#
```

STEP 08 – Self Study SNMP Commands

I. **snmpwalk** : Used to retrieve multiple SNMP objects. :
"snmpwalk -v2c -c public 10.0.1.2 system"

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmpwalk -v2c -c public 10.0.1.2 system
SNMPv2-MIB::sysDescr.0 = STRING: Linux mlb-dc1-centos9.csa.lk 5.14.0-565.el9.x86_64 #1 SMP PREEM
PT_DYNAMIC Fri Feb 7 18:47:11 UTC 2025 x86_64
SNMPv2-MIB::sysObjectID.0 = OID: NET-SNMP-MIB::netSnmpAgentOIDs.10
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (663307) 1:50:33.07
SNMPv2-MIB::sysContact.0 = STRING: admin@ndm.lk
SNMPv2-MIB::sysName.0 = STRING: mlb-dc1-centos9.csa.lk
SNMPv2-MIB::sysLocation.0 = STRING: "HYD, UM DataCenter"
SNMPv2-MIB::sysORLastChange.0 = Timeticks: (6) 0:00:00.06
SNMPv2-MIB::sysORID.1 = OID: SNMP-FRAMEWORK-MIB::snmpFrameworkMIBCompliance
SNMPv2-MIB::sysORID.2 = OID: SNMP-MPD-MIB::snmpMPDCompliance
SNMPv2-MIB::sysORID.3 = OID: SNMP-USER-BASED-SM-MIB::usmMIBCompliance
SNMPv2-MIB::sysORID.4 = OID: SNMPv2-MIB::snmpMIB
SNMPv2-MIB::sysORID.5 = OID: SNMP-VIEW-BASED-ACM-MIB::vacmBasicGroup
SNMPv2-MIB::sysORID.6 = OID: TCP-MIB::tcpMIB
SNMPv2-MIB::sysORID.7 = OID: UDP-MIB::udpMIB
SNMPv2-MIB::sysORID.8 = OID: IP-MIB::ip
SNMPv2-MIB::sysORID.9 = OID: SNMP-NOTIFICATION-MIB::snmpNotifyFullCompliance
SNMPv2-MIB::sysORID.10 = OID: NOTIFICATION-LOG-MIB::notificationLogMIB
SNMPv2-MIB::sysORDescr.1 = STRING: The SNMP Management Architecture MIB.
SNMPv2-MIB::sysORDescr.2 = STRING: The MIB for Message Processing and Dispatching.
SNMPv2-MIB::sysORDescr.3 = STRING: The management information definitions for the SNMP User-base
d Security Model.
SNMPv2-MIB::sysORDescr.4 = STRING: The MIB module for SNMPv2 entities
```

II. **snmptable** : Displays SNMP table data. : "snmptable -v2c -c public 10.0.1.2 ifTable"

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmptable -v2c -c public 10.0.1.2 ifTable
IF-MIB::ifTable: No entries
[root@mlb-dc1-centos9 svr1]#
```

III. **snmpset** : Used to modify SNMP variables. –
"snmpset -v2c -c private 127.0.0.1 system. sysName.0 s "MyServer""
"snmpget -v2c -c public 10.0.1.2 system.sysName.0S"

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmpset -v2c -c private 127.0.0.1 system.sysName.0 s "MyServer"
SNMPv2-MIB::sysName.0 = STRING: MyServer
[root@mlb-dc1-centos9 svr1]# snmpget -v2c -c public 10.0.1.2 system.sysName.0
SNMPv2-MIB::sysName.0 = STRING: MyServer
[root@mlb-dc1-centos9 svr1]#
```

IV. **snmptrap** : Used to send SNMP alerts. –
"snmptrap -v2c -c public 10.0.1.3 " SNMPv2-MIB::coldStart"

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmptrap -v2c -c public 10.0.1.3 '' SNMPv2-MIB::coldStart
[root@mlb-dc1-centos9 svr1]#
```